

ORIGINAL ARTICLE

A Clinical-Contextual Framework for Multimodal Tuberculosis Detection from Asynchronous CXR and Lung Sound Data using Lesion-Guided Pairing and Decision Fusion

Yudi Candra¹ | Roslidar Roslidar¹ | Mulkan Azhary² | Fitri Arnia¹ | Kahlil Muchtar¹

¹Department of Electrical and Computer Engineering, Universitas Syiah Kuala, Banda Aceh 23111, Indonesia

²Faculty of Medicine, Universitas Syiah Kuala, Banda Aceh 23111, Indonesia

Correspondence

Correspondence: Roslidar, Department of Electrical and Computer Engineering, Universitas Syiah Kuala.

Email: roslidar@usk.ac.id

Abstract

Tuberculosis (TB) remains a major global health challenge that requires diagnostic solutions capable of integrating structural and functional information from the respiratory system. This study proposes a novel lesion-guided multimodal framework that fuses chest X-ray (CXR) and lung sound data for TB detection, addressing the absence of synchronized datasets through a clinically grounded pairing mechanism. A YOLOv8-based segmentation model is employed to localize pulmonary lesions, which are then mapped to anatomically corresponding auscultation sites using a rib-guided quadrant extraction algorithm. This process generates a physiologically coherent paired dataset containing 450 CXR images and 2,700 lung-sound segments. A dual-branch architecture is developed, consisting of a ResNet50 encoder for visual features and a CRNN encoder for MFCC-based acoustic representations. Their probabilistic outputs are integrated using an optimized weighted decision-level fusion strategy, with weights derived from unimodal validation performance. Experimental results show that the proposed multimodal system outperforms both unimodal baselines, achieving 92% accuracy and an AUC of 0.98, compared with 0.95 (CXR-only) and 0.81 (audio-only). Qualitative Grad-CAM and saliency analyses further confirm that the model focuses on clinically meaningful radiographic lesions and crackle-like audio patterns. These findings demonstrate that the proposed lesion-guided multimodal framework offers a physiologically consistent and data-efficient approach for TB screening, highlighting its potential for future clinically deployable diagnostic systems.

KEY WORDS

Tuberculosis, Asynchronous Data Pairing, Decision-Level Fusion, Chest X-ray, Lung Sound

1 | INTRODUCTION

Tuberculosis (TB) remains one of the most persistent global health challenges despite being both preventable and curable. In 2024, an estimated 10.8 million people were infected and 1.25 million deaths occurred, reaffirming TB's position as one of the deadliest infectious diseases, with Indonesia contributing nearly 10% of the global burden^{1,2}. Clinically, TB is caused by *Mycobacterium tuberculosis*, an obligate aerobe with a strong predilection for the upper lung lobes where oxygen tension is highest³. This anatomical preference produces characteristic apical opacities on chest X-ray (CXR) and distinct respiratory abnormalities particularly crackles due to inflammation and tissue destruction⁴. The consistent correspondence between radiographic abnormalities and acoustic signatures provides a strong physiological foundation for multimodal diagnostic modeling.

Recent advances in deep learning have shown substantial potential in TB screening, especially through single-modality CXR models^{5,6}. Parallel progress in respiratory sound analysis further highlights the diagnostic value of acoustic signals^{7,8}, with studies such as Rajasekar et al. achieving 97% accuracy using cough audio⁹. However, CXR- and audio-based systems

Abbreviations: TB, tuberculosis; CXR, chest X-ray; AUC, area under the curve; MFCC, mel-frequency cepstral coefficients; CNN, convolutional neural network; CRNN, convolutional recurrent neural network; GRU, gated recurrent unit; MoE, mixture of experts; mAP, mean average precision; ROI, region of interest; PaO₂, partial pressure of oxygen; ICBHI, International Conference on Biomedical and Health Informatics.

remain limited when used independently, as each modality captures different physiological aspects of the same pathology. This fragmentation restricts diagnostic generalization and limits interpretability in real clinical settings.

Multimodal learning offers a promising solution by integrating heterogeneous data sources to exploit complementary and redundant features. Prior work supports this advantage: Zhou et al.¹⁰ reported a 12% AUROC gain when combining CXR and clinical text, while Kesarwani et al.¹¹ showed that fusing cough acoustics with patient metadata improves noise robustness. Ensemble-based strategies also demonstrate superior performance, as evidenced by Anand et al.¹². Despite these advances, no prior study has unified CXR and lung sound data within a single multimodal framework for TB detection—a significant gap given their inherent physiological complementarity.

Fusion strategy plays a crucial role in multimodal modeling. Early fusion suffers from scale and structural misalignment^{13,14}, while intermediate fusion enables richer cross-modal interactions through mechanisms such as attention¹⁴. Late fusion, which aggregates modality-specific decisions, is particularly effective for combining data with distinct statistical properties and noise characteristics^{15,16}.

Considering these challenges, this study aims to (1) resolve the dataset asynchrony between CXR and lung sound recordings and (2) develop a robust multimodal fusion framework for TB detection. To address the absence of synchronized multimodal datasets, we introduce a Lesion-Guided Smart Data Pairing mechanism that aligns TB-positive CXR images detected using YOLOv8 with clinically labeled crackle-containing lung-sound recordings. This anatomically informed pairing produces physiologically coherent CXR–audio samples. Modality-specific features are extracted using a fine-tuned ResNet50 for CXR images and a CRNN for lung sounds, followed by a grid-optimized weighted late-fusion strategy to improve interpretability and diagnostic balance.

The major contributions of this work are as follows:

1. A novel lesion-guided data pairing mechanism that synthetically synchronizes heterogeneous CXR and lung sound datasets based on anatomical logic.
2. A multimodal late-fusion architecture integrating CNN and CRNN encoders for independent yet complementary feature learning.
3. An optimized weighted ensemble strategy that enhances robustness and generalization across modalities.

Through this framework, the study provides a unified and interpretable multimodal approach for improving the reliability of AI-assisted tuberculosis screening systems.

2 | DATASET

This study adopts a multimodal framework built from two secondary medical datasets obtained from public repositories. A key challenge is the asynchronous and heterogeneous nature of the data: chest X-ray (CXR) images and lung sound recordings originate from different patient cohorts, making direct case-wise pairing infeasible. To overcome this limitation, a Lesion-Guided Smart Data Pairing mechanism was designed to ensure anatomical coherence between modalities during multimodal learning.

For the visual modality, thoracic images were sourced from the Shenzhen and Montgomery datasets^{17,18}, both consisting of PA-view radiographs labeled as Normal or Tuberculosis (TB) by expert radiologists. After removing samples with poor quality or inconsistent annotations, 450 images (225 Normal, 225 TB) were retained. All images were resized to 224×224 pixels and preprocessed using CLAHE and Gaussian blur to enhance lesion visibility and improve standardization.

For the auditory modality, the ICBHI 2017 Respiratory Sound Database^{19,20} was utilized. This dataset contains 920 clinically recorded lung-sound files annotated for crackles, wheezes, and normal breathing. A refined subset was created by selecting recordings with clear crackle or normal labels. Each recording was then segmented according to the official ICBHI annotation file, which provides the auscultation location and precise start–end timestamps of each respiratory event. Because a single recording typically contains 2–5 annotated events, this process yielded a total of 2,700 valid audio segments. After denoising via wavelet filtering and converting each segment into MFCC representations, these audio samples were subsequently aligned with CXR regions using lesion localization results to form a clinically coherent multimodal dataset.

3 | PROPOSED FRAMEWORK LESION-GUIDED SMART DATA PAIRING

The Lesion-Guided Smart Data Pairing Framework constitutes the first major component of the proposed multimodal tuberculosis (TB) detection system. This framework establishes a clinically grounded approach to constructing a coherent multimodal dataset

by leveraging lesion localization in chest X-ray (CXR) images as an anatomical reference for aligning corresponding lung sound data. Unlike conventional fusion methods that depend on co-collected patient records, it introduces a clinically informed synchronization strategy that integrates two heterogeneous modalities from independent repositories. As illustrated in Figure 1, the workflow comprises four main stages: (1) manual CXR annotation, (2) lesion detection using YOLOv8-seg, (3) anatomical mapping to standard auscultation sites, and (4) intelligent data pairing. The resulting dataset maintains both spatial and diagnostic consistency, where each CXR image is paired with six lung sound segments labeled as Normal or TB, serving as the foundation for subsequent multimodal feature extraction and fusion.

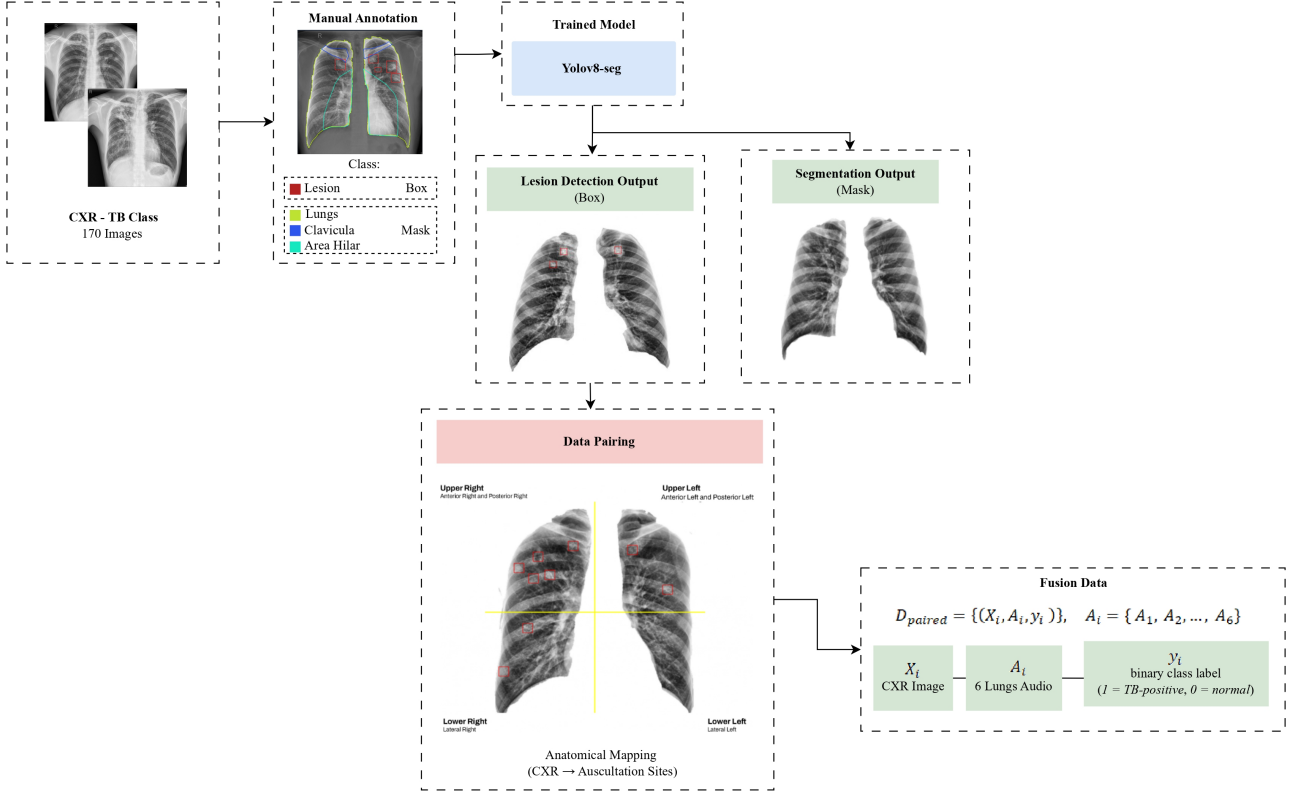


FIGURE 1 Workflow of the Lesion-Guided Smart Data Pairing Framework.

3.1 | Annotation and Lesion Localization

The annotation and lesion localization stage provides the foundational basis for developing the proposed multimodal tuberculosis (TB) detection framework. Its primary objective is to establish a clinically validated dataset that accurately reflects radiological manifestations of pulmonary TB. The process comprises two key components: (1) radiograph enhancement through standardized preprocessing and (2) manual annotation of pathological and anatomical structures to generate ground-truth references for automated lesion detection. Using the Roboflow platform, 170 TB-positive CXR images were annotated under pulmonologist supervision in accordance with WHO diagnostic standards. Four distinct classes were defined to guide lesion localization and contextual learning. The annotated classes were:

1. Lung Field, outlining the bilateral pulmonary region to define the primary analysis boundary.
2. Clavicle, identifying the collarbone structure, which often overlaps with upper-lobe lesions and may introduce false positives.
3. Hilar Region, marking the central thoracic area containing bronchi and blood vessels, critical for distinguishing vascular density from pathological opacity.
4. TB Lesion, delineating abnormal opacities such as infiltrates, consolidations, or cavitary lesions associated with active tuberculosis.

The original dataset consisted of 170 chest X-ray images from TB-positive cases, which were annotated in Roboflow and subsequently expanded through controlled augmentations: brightness adjustment (-20% to $+20\%$), blur (up to 1 px), and noise injection (up to 0.3% of pixels). After augmentation, the dataset was divided into 285 training images, 81 validation images, and 42 testing images, ensuring balanced representation of anatomical and pathological patterns across splits. All images were annotated using polygonal segmentation masks to achieve pixel-level delineation of anatomical and pathological structures. Each mask was stored in YOLOv8 segmentation format, containing class identifiers and normalized pixel coordinates. The finalized annotation set comprised 312 lesion regions and over 600 anatomical instances (lung fields, clavicles, and hila), capturing diverse morphologies and intensity variations. To ensure reliability, all annotations underwent a two-stage verification involving the primary annotator and a clinical expert, thereby eliminating anatomical overlaps and confirming pathological accuracy.

3.2 | YOLOv8-based Lesion Detection and Segmentation

This section presents the experimental evaluation of the YOLOv8-based lesion detection and segmentation framework, which forms the core component of the proposed Lesion-Guided Smart Data Pairing system. The evaluation encompasses model complexity, detection and segmentation accuracy, visual performance, and error analysis, aiming to identify the optimal configuration that balances diagnostic precision and computational efficiency for multimodal fusion. Four YOLOv8 segmentation variants, YOLOv8n-seg, YOLOv8s-seg, YOLOv8m-seg, and YOLOv8l-seg were fine-tuned using the annotated CXR dataset containing four segmentation classes: Lung Field, Clavicle, hilar region, and TB Lesion. Each model was trained for 100 epochs with a batch size of 8, a learning rate of 1×10^{-3} , and the Adam optimizer. Experiments were conducted on an NVIDIA RTX 3050 GPU (12 GB VRAM) with an input resolution of 640×640 pixels. Standard augmentation techniques, including random rotation and horizontal flipping, were employed to enhance model robustness. A unified evaluation framework was then applied to jointly assess architectural complexity (parameters, GFLOPs, and model size) and detection performance (Precision, Recall, and mAP), emphasizing the trade-off between computational cost and diagnostic reliability in medical AI systems.

TABLE 1 Comparison of YOLOv8 segmentation models.

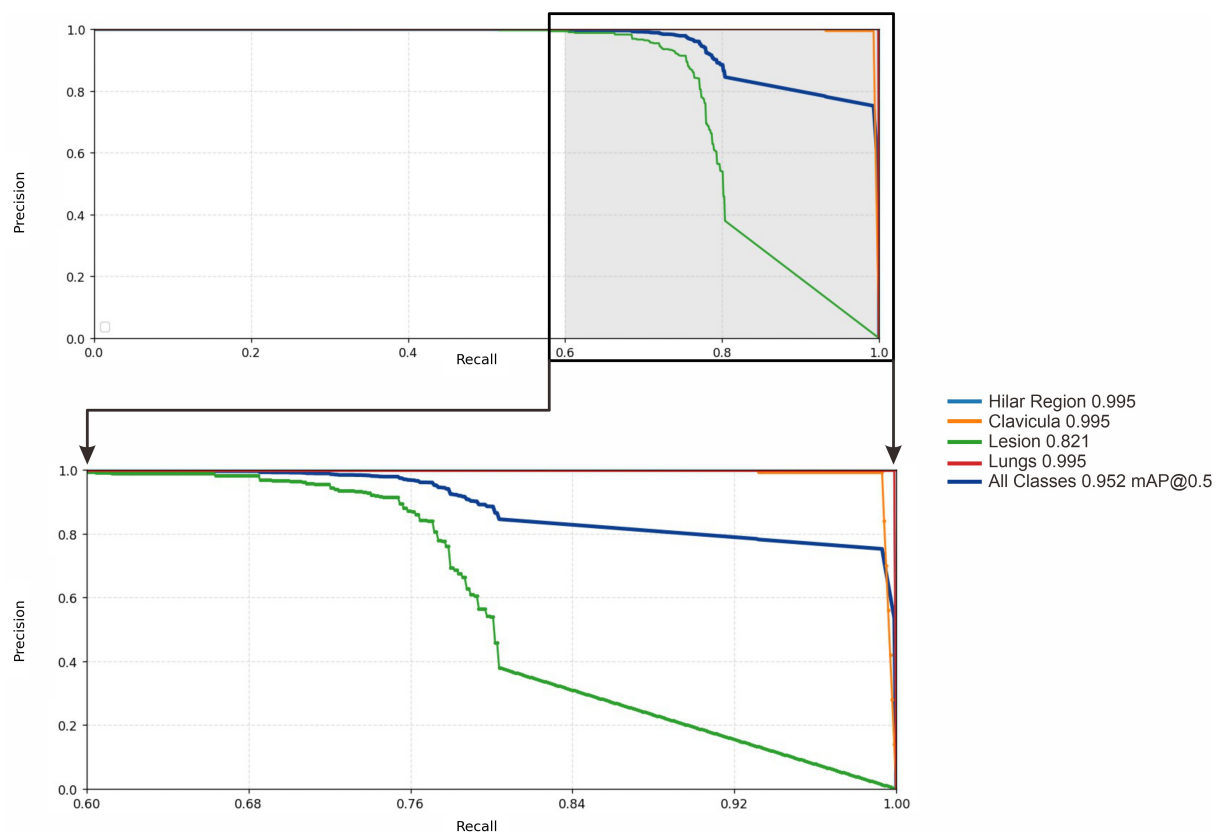
Model	Model Complexity			Model Performance			
	Params (M)	Model Size (MB)	Inference Time (ms/img)	Precision	Recall	mAP@50	mAP@50-95
YOLOv8n-seg	3.4	6.74	3.5	0.871	0.832	0.904	0.702
YOLOv8s-seg	11.2	22.8	5.1	0.889	0.857	0.923	0.739
YOLOv8m-seg	25.9	52.3	7.9	0.912	0.872	0.938	0.763
YOLOv8l-seg	43.7	88.1	11.6	0.926	0.889	0.951	0.784

The comparative results in Table 1 reveal a clear performance–complexity trend across the YOLOv8 segmentation variants. As model size increases, both detection accuracy and computational cost rise proportionally. The largest model, YOLOv8l-seg, achieves the strongest overall performance with Precision = 0.926, Recall = 0.889, and mAP@50 = 0.951, confirming its superior capability in lesion localization and anatomical boundary segmentation. However, this comes with substantial resource demands, including 43.7M parameters, 88.1 MB model size, and 11.6 ms/img inference time—limitations that reduce practicality for real-time or resource-constrained clinical environments. In contrast, YOLOv8m-seg presents an optimal balance between accuracy and efficiency: despite being significantly lighter (25.9M parameters, 52.3 MB) with a faster inference speed (7.9 ms/img), it retains competitive accuracy, achieving Precision = 0.912, Recall = 0.872, and mAP@50 = 0.938. Meanwhile, smaller variants such as YOLOv8n-seg and YOLOv8s-seg offer minimal computational load but exhibit noticeably lower mAP values (0.904 and 0.923, respectively), making them less reliable for fine-grained lesion detection. Overall, the results position YOLOv8m-seg as the most strategically balanced configuration, offering near-large-model accuracy with significantly reduced computational overhead, thus making it the most suitable backbone for robust and deployable clinical TB screening scenarios. To assess pixel-level precision, segmentation-specific mean Average Precision (mAP) metrics were calculated. The comparison between bounding-box and mask-based evaluation is shown in Table 2.

TABLE 2 Comparison of bounding-box and mask-based segmentation performance across YOLOv8 variants.

Model	Model Performance			
	mAP@50 (Box)	mAP@50–95 (Box)	mAP@50 (Mask)	mAP@50–95 (Mask)
YOLOv8n-seg	0.904	0.702	0.881	0.672
YOLOv8s-seg	0.923	0.739	0.899	0.714
YOLOv8m-seg	0.938	0.763	0.917	0.735
YOLOv8l-seg	0.951	0.784	0.932	0.758

Mask-based mAP values are slightly lower due to the stricter pixel-overlap criterion, yet the results indicate that the segmentation branch of YOLOv8m-seg accurately delineates lesion morphology and remains well aligned with expert annotations. To further assess the reliability of the detection component, a class-aggregated Precision–Recall (PR) curve was generated (Figure 2). The PR curve captures the fundamental trade-off between precision and recall across all segmentation classes and is particularly informative for medical object detection under class imbalance conditions. The TB lesion class achieved an F1-score exceeding 0.82, while anatomical structures such as the lung field, clavicle, and hilar region reached near-saturated performance levels ($F1 > 0.95$). The model also attained a mean Average Precision of 0.952 (mAP@0.5), demonstrating strong generalization across diverse lesion shapes and sizes. Overall, these findings establish YOLOv8m-seg as a robust and reliable detector within the proposed Lesion-Guided Smart Data Pairing framework.

**FIGURE 2** Precision–Recall curve of the model accompanied by a magnified view of the high-precision region.

In addition to the quantitative metrics, the confusion matrix in Figure 3 provides a detailed quantification of the YOLOv8m-seg model's class-wise segmentation performance. The model demonstrates strong diagonal dominance across all classes, with the Lesion category achieving the highest correct detection count of 374 samples, reflecting robust sensitivity to pathological regions. Anatomical structures also exhibit high accuracy: the hilar region class records 162 correct predictions, Clavicle 159, and Lungs 162, indicating that the model reliably distinguishes major thoracic components despite variations in rib density and overlapping soft tissues. Misclassifications occur primarily in anatomically adjacent regions, such as 29 lesion samples mislabeled as

background, and 3 lung samples misidentified as background, reflecting expected boundary-level ambiguity in low-contrast areas. Additionally, 119 lesion instances were predicted as background, suggesting that subtle or small lesions remain challenging under certain radiographic conditions. Overall, these quantitative patterns underscore the model's strong discriminative capability across anatomical and pathological categories while highlighting clinically plausible sources of segmentation error.

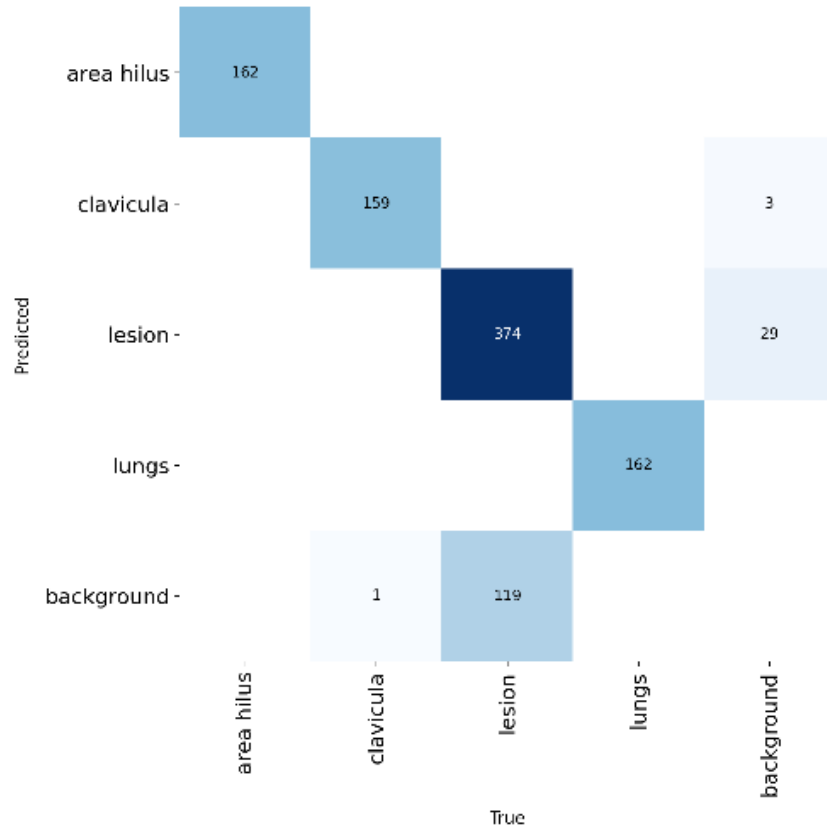


FIGURE 3 Confusion matrix of the YOLOv8m-seg model illustrating class-wise detection accuracy for anatomical and pathological categories.

The results reveal a clear accuracy–efficiency trade-off across YOLOv8 model scales. Larger models (YOLOv8l-seg) deliver the highest detection accuracy but require significantly greater computational resources, which may limit real-time deployment. Smaller variants (YOLOv8n-seg and YOLOv8s-seg) offer faster inference but lower sensitivity to small or diffuse lesions. The YOLOv8m-seg variant demonstrates the optimal balance between accuracy and efficiency, achieving $mAP@50 = 0.938$, Precision = 0.912, and an inference time of 7.9 ms/image. These results justify its selection as the backbone detector for the subsequent Lesion-Guided Smart Data Pairing mechanism, enabling anatomically informed multimodal alignment between CXR and lung sound data. In addition to lesion localization, the YOLOv8m-seg model also performs lung-field segmentation to remove regions outside the pulmonary area, ensuring that only anatomically valid lesions within the lung boundaries are retained for the subsequent multimodal pairing process.

3.3 | Intelligent Data Pairing Mechanism

To address the lack of synchronized CXR–audio datasets, an Intelligent Data Pairing mechanism was developed to establish anatomically meaningful correspondences between radiographic lesions and region-specific lung sounds. This ensures that each paired sample reflects a clinically plausible relationship between visual abnormalities and their acoustic manifestations.

The YOLOv8m-seg model detects four anatomical and pathological structures: lung field, clavícula, hilar region, and TB lesion. The lung-field segmentation mask is used as an anatomical constraint so that only lesions located within the thoracic

cavity are retained. For each valid lesion, its geometric centroid $(\bar{x}, \bar{y})$ and area A are computed using standard spatial moments²¹:

$$\bar{x} = \frac{1}{A} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{A} \sum_{i=1}^n y_i. \quad (1)$$

Traditional quadrant definitions such as the Modified Chrispin–Norman scoring system require manual identification of the hilar boundary, which is subjective and varies across patients²². To obtain a reproducible horizontal separator, this study employs a rib-based approach by locating the posterior fourth and fifth ribs. The midpoint between these ribs provides a consistent boundary for separating upper and lower lung zones.

A reference 20×20 rib-overlap patch is first selected to represent the characteristic intensity pattern at rib intersections. Candidate patches across the CXR image are then compared with this reference using the Euclidean distance:

$$d_{x,y} = \|\mathbf{s}_{x,y} - \mathbf{s}_r\|_2. \quad (2)$$

In this formulation, $\mathbf{s}_{x,y}$ denotes the intensity descriptor of the patch centered at (x, y) , $\mathbf{s}_r$ is the descriptor of the reference rib patch, and $d_{x,y}$ represents their similarity measure, with smaller values indicating closer resemblance. The most similar patches are subsequently clustered to obtain landmark points that serve as rib-counting indicators rather than true anatomical rib centers. The posterior fourth–fifth rib pair is then used to derive the horizontal separator y_h , while the vertical midline x_v is obtained from the mediastinal center. Together, x_v and y_h partition the lung field into four anatomical quadrants (UR, UL, LR, and LL). The complete procedure is summarized in Algorithm 1.

Algorithm 1 Rib-Based Quadrant Extraction for Anatomically Guided Cross-Modal Alignment

Require: Chest X-ray image I_0

Ensure: Quadrant assignment Q for each lesion

```

1: Extract reference rib patch and compute its descriptor
2: for each  $20 \times 20$  patch in  $I_0$  do
3:   Compute descriptor and similarity score  $d_{x,y}$ 
4: end for
5: Retain the top 1% rib-like patches and filter by lateral ROI
6: Cluster remaining points to estimate rib centers
7: Identify the 4th–5th rib pair and compute horizontal separator  $y_h$ 
8: Determine vertical midline  $x_v = W/2$ 
9: for each lesion centroid  $(x_\ell, y_\ell)$  do
10:   Assign quadrant  $Q_\ell$  based on  $(x_v, y_h)$ 
11: end for
12: return all quadrant labels  $Q$ 

```

Figure 4 illustrates the process of dividing each chest X-ray into four anatomical quadrants using the vertical and horizontal reference lines. These quadrants serve as spatial anchors for localizing detected lesions (highlighted with red bounding boxes) within specific lung regions. The green rib landmark points refine the alignment and separation of the left and right lungs, ensuring consistent anatomical partitioning. Each detected lesion is assigned to a quadrant based on its centroid position.

To establish a physiologically meaningful correspondence between CXR lesions and lung sounds, each quadrant is then mapped to the six auscultation sites defined in the ICBHI 2017 Respiratory Sound Dataset (AR, PR, AL, PL, LR, LL). The mapping follows standard clinical auscultation practice: each upper quadrant corresponds to two sites (anterior and posterior), while each lower quadrant corresponds to one site, resulting in six region-specific acoustic inputs per CXR. In this study, only the normal and crackle labels from ICBHI are used. A quadrant is assigned a crackle label if at least one lesion lies within it; otherwise, it is labeled normal. This rule-based alignment ensures a physiologically coherent link between radiographic abnormalities and their expected acoustic manifestations.

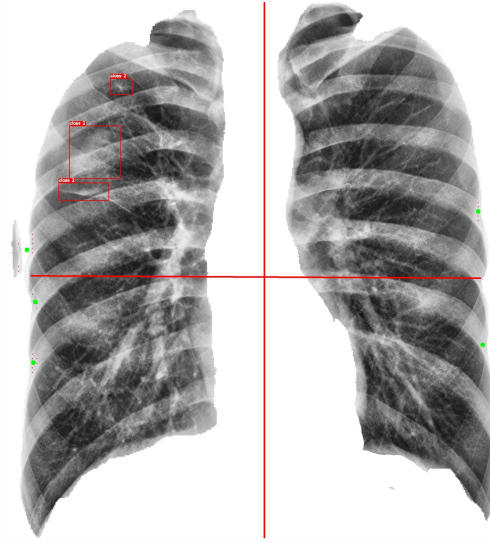


FIGURE 4 Example of Quadrant Separation on Chest Radiograph.

Figure 5 illustrates the distribution of 242 detected lesions across 42 test images. The counts show a clear predominance in the upper lung regions, with 109 lesions located in the upper left quadrant and 99 lesions in the upper right quadrant. In contrast, the lower zones contain substantially fewer lesions, with 21 in the lower right and 13 in the lower left quadrant. This concentration of lesions in the upper lung fields aligns with the well-established pathophysiology of pulmonary tuberculosis, which typically manifests more prominently in the apical and upper regions of the lungs due to higher oxygen tension that favors *Mycobacterium tuberculosis* proliferation³.

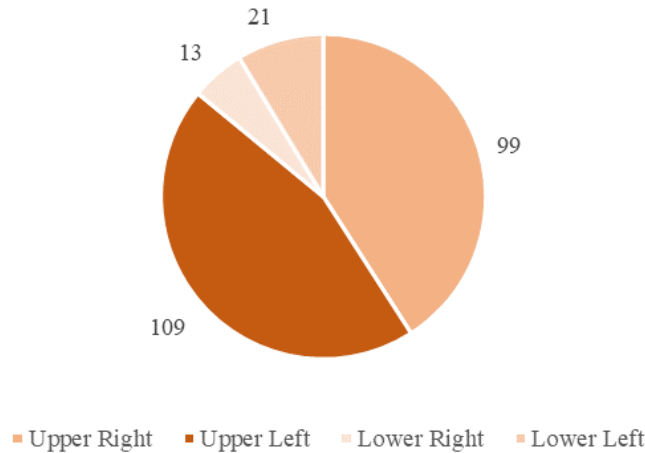


FIGURE 5 Spatial Distribution of Detected Lesions in the Test Dataset.

Using this procedure, a total of 450 CXR images were paired with six auscultation recordings each, resulting in a multimodal dataset comprising 3,150 paired samples. Equation 3 formulated the pairing.

$$D_{\text{paired}} = (X_i, A_i, y_i), \quad A_i = A_1, A_2, \dots, A_6. \quad (3)$$

In this representation, X_i denotes the i -th CXR image, A_i represents the corresponding set of six auscultation segments, and y_i is the associated class label. To strictly prevent cross-subset data leakage, the splitting process was performed at the synthetic-patient level rather than at the individual sample level. Each CXR together with its six paired audio segments was considered as belonging to a single synthetic patient ID; critically, all six auscultation segments assigned to a synthetic patient were sourced

from the *same original audio subject*, ensuring that no synthetic patient mixes lung-sound segments from different individuals. Likewise, when an original ICBHI audio recording contained multiple annotated events, all resulting segments were kept together within the same synthetic-patient grouping and were never distributed across different IDs. Therefore, once a synthetic patient (e.g., ID 039) was assigned to the training set, none of its associated CXR or audio segments appeared in the validation or test sets. This subject-level separation ensures statistical independence between splits and avoids unintended information sharing across subsets. The resulting dataset was then partitioned into training, validation, and test subsets using an 80:10:10 stratified split, as summarized in Table 3. Overall, this mechanism integrates lesion detection, anatomical boundary estimation, rib-based quadrant separation, and rule-based audio assignment to construct a clinically coherent multimodal dataset for subsequent fusion-based TB classification.

TABLE 3 Summary of the Final Multimodal Dataset Generated Through the Smart Data Pairing Process.

Subset	CXR Images	Audio Segments	Total Samples
Train	360	2160	2520
Validation	45	270	315
Test	45	270	315
Total	450	2700	3150

4 | DECISION-LEVEL FUSION

The Decision-Level Fusion Framework constitutes the core computational stage of the proposed multimodal tuberculosis (TB) detection system. As illustrated in Figure 6, the framework comprises two modality-specific deep learning branches: a convolutional neural network (CNN) for visual CXR analysis and a convolutional recurrent neural network (CRNN) for audio-based respiratory sound interpretation. Each branch is independently optimized to extract discriminative features within its modality domain. The subsequent decision-level fusion module combines their probabilistic outputs using a weighted averaging ensemble, enabling robust and interpretable multimodal integration. This hierarchical organization ensures that each modality contributes complementary diagnostic evidence, thus enhancing overall system reliability and reducing modality-specific bias.

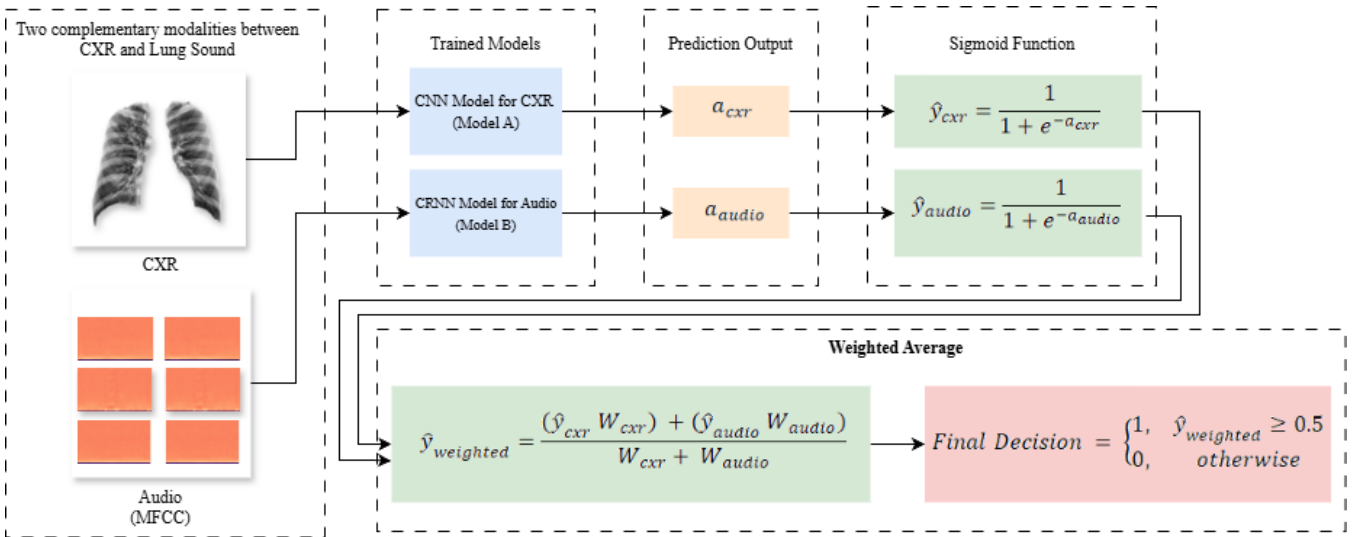


FIGURE 6 Structure of the dual-branch CNN-CRNN architecture and weighted fusion mechanism used for multimodal.

4.1 | Modality-Specific Feature Extraction

The feature extraction stage focuses on learning modality-dependent representations from both visual and acoustic data. For the CXR branch, a pretrained ResNet50 convolutional backbone was employed owing to its proven capability in medical image representation. The model was fine-tuned on the curated CXR dataset described in Section II, with the final classification

head modified to output a single scalar logit corresponding to TB probability. The output activation was handled internally via the BCEWithLogitsLoss function, ensuring numerical stability during optimization. To enhance generalization, extensive augmentation was applied, including random rotation ($\pm 10^\circ$), horizontal flipping, and intensity scaling. Each image was resized to 224×224 pixels and normalized to ImageNet statistics before being fed into the network.

For the audio branch, a Convolutional Recurrent Neural Network (CRNN) was implemented to jointly model the spectral and temporal characteristics of respiratory sounds. Each lung-sound recording was transformed into a Mel-Frequency Cepstral Coefficient (MFCC) spectrogram that preserves both frequency distribution and time-varying patterns. The CRNN consists of stacked 2-D convolutional layers for local spectral extraction followed by bidirectional GRU layers to capture long-term temporal dependencies across respiratory cycles. A fully connected layer then produces a single scalar logit analogous to the CXR branch.

where $X \in \mathbb{R}^{224 \times 224 \times 3}$ denotes the pre-processed chest X-ray image, and $S \in \mathbb{R}^{T \times F}$ represents the MFCC-based spectrogram of the lung sound, with T and F corresponding to the number of temporal frames and frequency coefficients, respectively. In this study, each MFCC spectrogram had a fixed dimension of $T = 395$ and $F = 40$, obtained using a 40 ms frame size with 20 ms overlap.

The CXR branch, implemented using a fine-tuned ResNet50 backbone, extracts high-level spatial representations from localized pulmonary lesions. In parallel, the audio branch, implemented as a CRNN, captures both local spectral characteristics and long-range temporal dependencies through stacked 2-D convolutional and bidirectional GRU layers. Each branch outputs a scalar logit a_{cxr} for the CXR stream and a_{audio} for the audio stream—which is subsequently transformed into a probabilistic prediction using the sigmoid activation²³:

$$\hat{y} = \sigma(a) = \frac{1}{1 + e^{-a}}. \quad (4)$$

In this expression, a denotes the input logit, $\sigma(\cdot)$ is the sigmoid function, and $\hat{y}$ represents the resulting probability of tuberculosis presence. These modality-specific probability estimates serve as the inputs to the subsequent fusion stage.

4.2 | Late Fusion Mechanism

To integrate the probabilistic outputs from both modalities, this study evaluates two decision-level fusion strategies, beginning with a static *Weighted Averaging* ensemble. Following the classical formulation in *Ensemble Methods: Foundations and Algorithms*²⁴, the fused decision function is expressed as:

$$H(x) = \sum_{i=1}^T w_i h_i(x), \quad (5)$$

where $h_i(x)$ denotes the output of the i -th learner and w_i is its non-negative weight satisfying $\sum_{i=1}^T w_i = 1$. In the proposed multimodal setting ($T = 2$), the ensemble combines the probabilistic predictions of the CXR and audio branches, denoted as $\hat{y}_{\text{cxr}}$ and $\hat{y}_{\text{audio}}$, respectively. The weighted fusion output is therefore given by:

$$\hat{y}_{\text{weighted}} = W_{\text{cxr}} \hat{y}_{\text{cxr}} + W_{\text{audio}} \hat{y}_{\text{audio}}, \quad (6)$$

where W_{cxr} and W_{audio} are fusion weights selected through grid-search on the validation set to maximize the AUC. This approach ensures a simple and interpretable combination, with each modality contributing a fixed proportion to the final prediction.

In contrast, the second strategy employs a dynamic *Mixture-of-Experts* (MoE) mechanism in which the fusion weights are generated adaptively for each input. The MoE output is defined as²⁴:

$$H_{\text{moe}}(x) = \sum_{i=1}^T g_i(x) h_i(x), \quad (7)$$

where $g_i(x)$ is a learnable gating function producing sample-specific weights that satisfy $g_i(x) \geq 0$ and $\sum_{i=1}^T g_i(x) = 1$. This enables the fusion module to emphasize the CXR stream when radiographic abnormalities are more informative, or the audio stream when acoustic patterns dominate. Although more flexible than static fusion, the MoE approach introduces greater model complexity and reduced interpretability.

The final binary TB decision is obtained by applying a fixed threshold to the fused probability as formulated in equation 8.

$$\text{Final Decision} = \begin{cases} 1, & \hat{y} \geq 0.5, \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

Performance evaluation was conducted using standard classification metrics derived from the confusion matrix. Let TP , TN , FP , and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively. The evaluation metrics are defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}, \quad (9)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (10)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (11)$$

$$F1\text{-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (12)$$

where Accuracy measures overall correctness, Precision reflects the proportion of predicted positives that are true positives as TB, Recall quantifies the model's ability to detect actual positive cases of TB, and the F1-Score provides a harmonic balance between Precision and Recall.

4.3 | Experimental Setup

All experiments were performed on an NVIDIA RTX 3050 GPU using Python 3.11.9 and PyTorch 2.2.1. The multimodal dataset, constructed via the Lesion-Guided Smart Pairing strategy, yielded 3,150 CXR–audio pairs from 450 scans and was split into 80% training, 10% validation, and 10% testing.

Each batch contained eight paired samples, where CXR images were prepared as $3 \times 224 \times 224$ tensors and the audio stream consisted of six MFCC spectrograms of size 40×395 from the six auscultation sites. Both unimodal models—ResNet50 for CXR and a CRNN for audio—were trained for 100 epochs using Adam, BCE loss, an initial learning rate of 1×10^{-4} with scheduled decay, early stopping, and a fixed random seed. Data augmentation was applied independently to each modality, and class imbalance was handled using a `WeightedRandomSampler`.

4.4 | Result and Discussion

4.4.1 | Unimodal Baseline Performance

To evaluate the diagnostic capability of each modality before multimodal fusion, two unimodal baseline models were trained under identical experimental conditions. Model A (CXR-only) employed a fine-tuned ResNet50 for chest X-ray classification, while Model B (Audio-only) used a Convolutional Recurrent Neural Network (CRNN) trained on MFCC spectrograms of lung sounds. Both models were trained for 100 epochs using the Adam optimizer and Binary Cross-Entropy (BCE) loss, with early stopping and modality-specific data augmentation to enhance generalization.

As shown in Table 4, the CXR model achieved superior performance (Accuracy = 0.90, Precision = 0.91, Recall = 0.90, F1 = 0.90, AUC = 0.95) compared to the audio model (Accuracy = 0.73, Precision = 0.74, Recall = 0.73, F1 = 0.73, AUC = 0.81). These results indicate that radiographic features provide stronger and more stable diagnostic cues for TB detection, whereas audio-based features serve as complementary but less consistent indicators when used in isolation.

TABLE 4 Quantitative performance of unimodal baseline models.

Model	Accuracy	Precision	Recall	F1-Score	AUC
Model A (CXR)	0.90	0.91	0.90	0.90	0.95
Model B (Audio)	0.73	0.74	0.73	0.73	0.81

The training history illustrated in Figure 7 shows a smooth and monotonic decrease in training and validation losses for the CXR-only model, indicating strong convergence behavior. Its accuracy curve rapidly stabilized above 0.95 after approximately 30 epochs, with AUC approaching 1.00, signifying near-optimal discriminative separation between normal and TB cases. Conversely, the audio-only model exhibited greater fluctuation during training, with noticeable oscillations in both accuracy and loss curves. Despite achieving a training accuracy above 0.90, its validation accuracy plateaued around 0.80, suggesting moderate generalization and higher sensitivity to inter-patient variability and environmental acoustic noise.

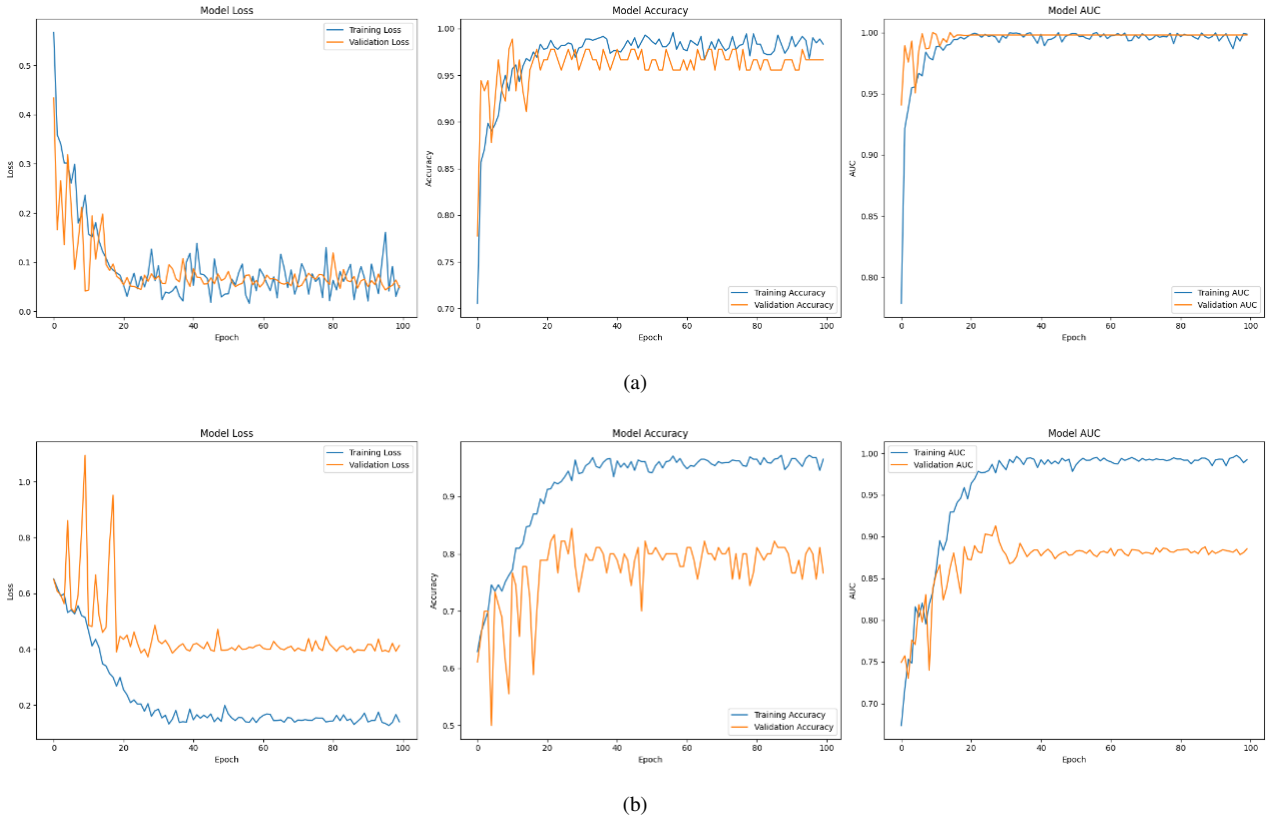


FIGURE 7 Training history of unimodal baseline models: (a) CXR-only model and (b) Audio-only model.

The confusion matrices presented in Figure 8 further confirm these findings. Model A (CXR-only) correctly classified 45 of 46 normal samples and 36 of 44 TB-positive samples, yielding strong diagonal dominance and minimal false negatives. This performance reflects the model's ability to identify subtle radiographic lesions in the apical regions of the lungs. In contrast, Model B (Audio-only) exhibited higher misclassification rates, particularly for TB cases, where 16 of 44 samples were incorrectly labeled as normal. This result highlights the inherent variability and overlapping acoustic patterns between pathological and healthy lung sounds, which pose additional challenges for unimodal audio-based diagnosis.

4.4.2 | Multimodal Fusion Performance

Building upon the unimodal baselines, several decision-level fusion strategies were implemented to integrate the probabilistic predictions from both the CXR and audio classifiers. Among these, the *Weighted Averaging Fusion* demonstrated the best overall performance. The fusion weights were empirically optimized based on the validation AUC of each unimodal model, yielding $W_{\text{cxr}} = 0.51$ and $W_{\text{audio}} = 0.49$ (Validation AUC = 0.98). This relatively higher validation AUC occurs because the fusion is evaluated on the paired multimodal validation subset, which exhibits lower variance and a more balanced lesion–normal distribution than the unimodal validation splits, resulting in a smoother ROC curve. This near-balanced weighting reflects the complementary reliability of both modalities, while slightly favoring the CXR stream due to its higher discriminative consistency. The reliability-based fusion scheme maintains interpretability and computational efficiency, consistent with ensemble learning principles for heterogeneous data integration.

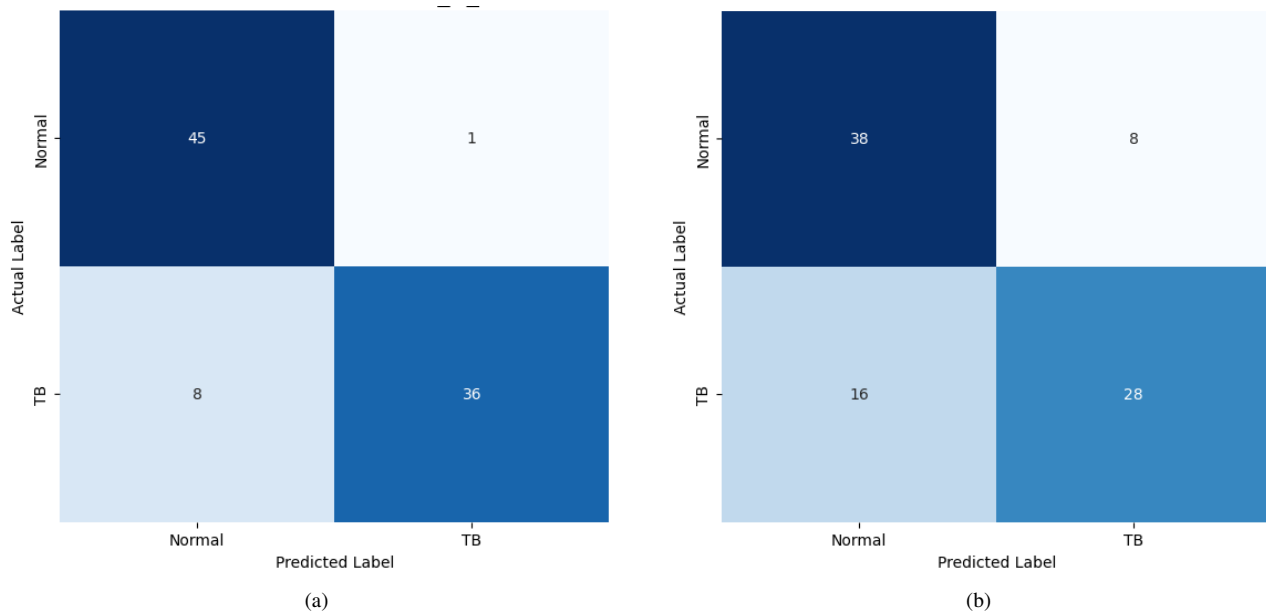


FIGURE 8 Confusion matrices of unimodal models: (a) Model A (CXR) (b) Model B (Audio).

As summarized in Table 5, the Weighted Average Fusion achieved the highest test performance, reaching 0.92 accuracy, 0.93 precision, 0.92 recall, 0.92 F1-score, and an AUC of 0.98. These metrics represent a substantial improvement compared to the audio-only classifier (accuracy = 0.73, AUC = 0.81) and a moderate enhancement over the CXR-only baseline (accuracy = 0.90, AUC = 0.95). The Max Fusion method, which selects the higher confidence score per sample, also achieved competitive results (accuracy = 0.89, AUC = 0.91) but exhibited a slightly higher false-positive rate. In contrast, the Mixture of Experts (ME) dynamic fusion model yielded the lowest stability, achieving accuracy = 0.79 and AUC = 0.90, indicating that the learned gating function requires further tuning or larger-scale training data to generalize effectively.

TABLE 5 Performance comparison between unimodal and fused models.

Method	Accuracy	Precision	Recall	F1-Score	AUC
CXR-Only	0.90	0.91	0.90	0.90	0.95
Audio-Only	0.73	0.74	0.73	0.73	0.81
Weighted Average Fusion	0.92	0.93	0.92	0.92	0.98
Max Fusion	0.89	0.90	0.89	0.89	0.91
Mixture of Experts (ME)	0.79	0.79	0.79	0.79	0.90

The confusion matrices in Figure 9(a–c) reveal distinct error characteristics for each fusion variant. The Weighted Average Fusion achieved the most stable classification, correctly identifying all 46 Normal cases and 37 out of 44 TB cases, with 7 false negatives and no false positives, indicating a strong balance between sensitivity and specificity. The Max Fusion method favored sensitivity, correctly classifying 43 of 44 TB cases but misclassifying 9 Normal samples. Meanwhile, the Mixture of Experts exhibited inconsistent weighting behavior, misclassifying 19 samples (8 Normal and 11 TB), reflecting limited robustness in this configuration.

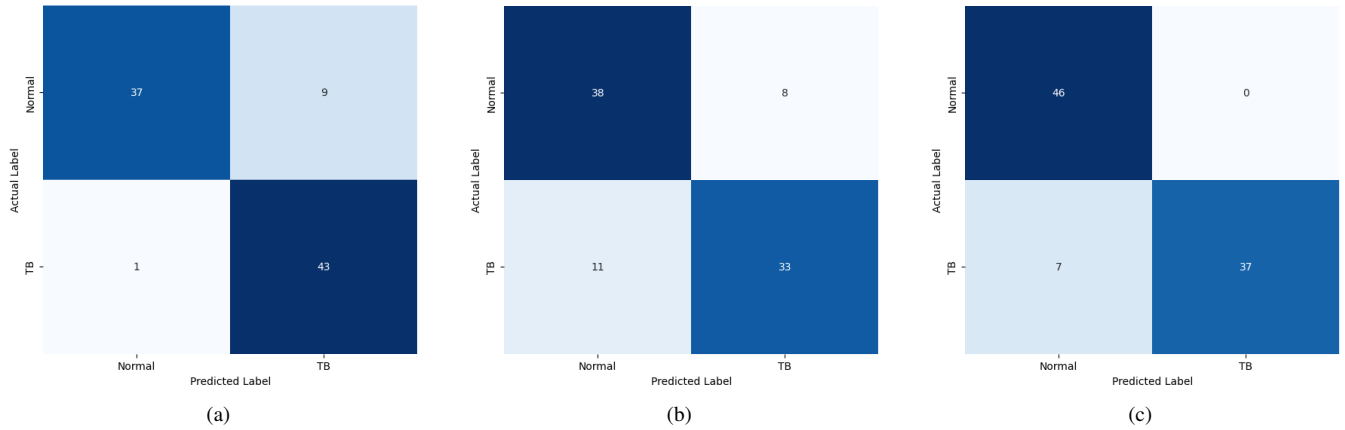


FIGURE 9 Confusion matrices arranged as (a) Max Fusion, (b) Mixture-of-Experts, and (c) Weighted Average Fusion.

The Weighted Average Fusion produced the steepest ROC trajectory and the highest AUC (0.98), outperforming both unimodal models (CXR-only = 0.95, Audio-only = 0.81) and the alternative fusion schemes (Max Fusion = 0.91, Mixture Experts = 0.90). The improved curve shape near the upper-left region indicates enhanced sensitivity across varying thresholds, confirming the successful integration of complementary radiographic and acoustic features.

Collectively, these results affirm that the proposed Lesion-Guided Smart Pairing and Weighted Decision-Level Fusion ($W_{\text{cxr}} = 0.51$, $W_{\text{audio}} = 0.49$) significantly enhance tuberculosis detection performance. By combining the structural insights of chest X-rays with the functional cues of lung sounds, the multimodal system achieves a more robust and clinically interpretable diagnostic outcome, demonstrating the effectiveness of multimodal ensemble learning for AI-assisted TB screening.

4.4.3 | Qualitative Analysis

To further analyze the decision behavior of the proposed multimodal tuberculosis detection system, we generated qualitative interpretability visualizations for both modalities. For the CXR branch, we used Gradient-weighted Class Activation Mapping (Grad-CAM), which backpropagates the gradient of the target class score to the last convolutional feature maps and computes a weighted combination of these maps. The resulting heatmap highlights spatial regions whose activations contribute most strongly to the predicted class, thus providing a class-discriminative localization of salient lung structures. This heatmap is then upsampled and overlaid on the original CXR image, allowing visual inspection of whether the model focuses on clinically plausible areas such as parenchymal opacities, cavitations, or upper-lobe lesions.

For the MFCC-based audio branch, we applied saliency mapping by computing the gradient of the output score with respect to the input spectrogram. The magnitude of these gradients indicates how sensitive the prediction is to small perturbations at each time–frequency bin, thereby revealing the temporal segments and frequency bands that most influence the model’s decision. When superimposed on the MFCC representation, these saliency maps highlight candidate adventitious events (e.g., crackle-like bursts) associated with abnormal lung sounds.

As illustrated in Figures 10(a) and 10(b), the normal case exhibits low-intensity and spatially diffuse Grad-CAM responses across the lung fields, together with weak and scattered saliency patterns on the MFCC spectrogram. This combination suggests the absence of distinctive radiographic or acoustic abnormalities and indicates that the model does not latch onto spurious localized artefacts. In contrast, the TB-positive example shows strong, localized Grad-CAM activations concentrated in the upper lung regions—consistent with typical tuberculosis lesion sites—as well as clearly delineated high-saliency intervals in the audio domain that correspond to crackle-like events. The spatial–temporal alignment of these discriminative regions across modalities demonstrates coherent multimodal reasoning and confirms that the fusion model is leveraging physiologically meaningful cues rather than dataset biases or noise, thereby reinforcing the reliability and clinical plausibility of its predictions.

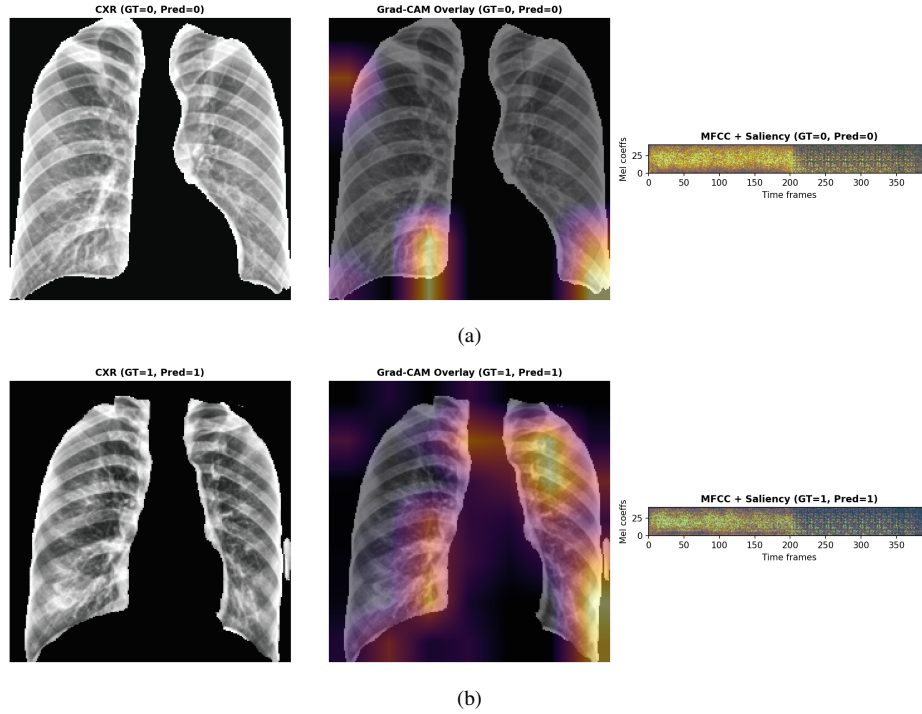


FIGURE 10 Representative qualitative visualizations for correctly classified test samples: (a) normal case and (b) tuberculosis-positive case

4.4.4 | Summary

The experimental results demonstrated that the proposed multimodal tuberculosis detection framework, combining Lesion-Guided Smart Data Pairing and Weighted Decision-Level Fusion, significantly outperformed unimodal baselines. The CXR model (AUC = 0.95) provided strong structural diagnostic cues, while the audio model (AUC = 0.81) contributed complementary functional information. Their weighted fusion (CXR = 0.51, Audio = 0.49) achieved the best overall performance with 0.92 accuracy and 0.98 AUC, confirming the effectiveness of multimodal integration. Qualitative visualizations further revealed that the model's attention aligned with clinically relevant regions and acoustic events, validating its interpretability and physiological consistency.

5 | CONCLUSIONS AND FUTURE WORK

5.1 | Conclusion

This study proposed a clinically grounded multimodal deep learning framework for tuberculosis detection by integrating chest X-ray and lung sound data through Lesion-Guided Smart Pairing and Weighted Decision-Level Fusion. The experimental findings confirmed that combining structural and functional features substantially improved diagnostic reliability, achieving an AUC of 0.98 with high interpretability. The fusion mechanism effectively balanced contributions from both modalities (CXR = 0.51, Audio = 0.49), enhancing robustness while maintaining clinical coherence. Overall, the proposed approach successfully bridged the gap between heterogeneous modalities and demonstrated the potential of interpretable AI systems for accurate, data-efficient TB screening.

5.2 | Future Work

Future research should focus on enhancing dataset quality and model generalization by employing primary multimodal datasets collected from the same patients to ensure better physiological alignment between CXR and audio data. Improving the accuracy of the audio branch using transformer-based encoders, attention mechanisms, or noise-robust feature extraction is also essential. Additionally, incorporating other clinical modalities such as cough analysis or textual patient data could further strengthen

diagnostic context. A complementary research direction is the systematic analysis of misclassified samples—particularly false positives and false negatives—to identify underlying factors that lead to erroneous predictions in both CXR and audio branches, such as atypical lesion morphology, poor signal quality, or label noise. Insights from such error analysis may guide targeted model refinement and data curation. Finally, model optimization for edge or mobile deployment and clinical validation with real-world cases will be vital steps toward realizing a deployable and trustworthy AI-assisted tuberculosis screening system.

DATA AVAILABILITY STATEMENT

The original chest X-ray and respiratory sound datasets analyzed in this study are publicly available from the Shenzhen and Montgomery TB collections and the ICBHI 2017 Respiratory Sound Database respectively, as cited in the manuscript. The processed lesion annotations and the paired multimodal dataset are available from the corresponding author on reasonable request due to privacy and ethical restrictions.

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CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

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